REGISTRATION CENTRE NUMBER	CENTRE NAME	
CANDIDATE'S FULL NAMES		
CANDIDATE'S IDENTIFICATION NUMBER	SUBJECT CODE **0515**	PAPER NUMBER **2**

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GENERAL CERTIFICATE OF EDUCATION BOARD

General Certificate of Education Examination

ORDINARY LEVEL

SUBJECT TITLE **CHEMISTRY**	SUBJECT CODE **0515**	PAPER NUMBER **2**
	EXAMINATION DATE: **JUNE 2026**	

Duration: Two and a Half Hours

Enter the information required in the boxes above.

This paper is arranged in three **SECTIONS, A, B and C.**

*Section A: Answer **all 5** questions. You will be graded for the best 4 answers.*

*Section B: Answer **all 2** questions.*

*Section C: Answer **2 questions** out of 3.*

In calculations, you are advised to show all the steps in your working, giving your answer at each stage. Calculators are allowed.

You are reminded of the necessity for good English and orderly presentation in your answers.

USEFUL DATA: 1 Faraday = 96000 Coulombs.
Molar volume of a gas at r.t.p. = 24000 cm^3,
Specific Heat Capacity of water = 4.2 J/g/°C
Avogadro Number = 6.02×10^{23Ca}
O°C = 273K

Relative Atomic Masses	
Hydrogen	1
Carbon	12
Oxygen	16
Chlorine	35,5
Silver	108

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SCORE

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SECTION A

ANSWER ALL 5 questions. You will be graded for the best 4 answers.

1) This question concerns the elements $^{24}_{12}X$ and $^{18}_{8}Y$ where X and Y are not the usual symbols of the elements.

a) Write the electronic configuration of X and Y

X.....

Y.....

(2 marks)

b) To which group and period of the Periodic Table does X belong?

Group ..... Period .....

(2 marks)

c) (i) What does the figure 24 in $^{24}_{12}X$ represent?.....

(ii) How many protons are found in Y?.....

(iii) How many neutrons are found in X?.....

(3 marks)

d) Using electron dot and cross diagram only, show the bonding in the compound formed between X and Y.

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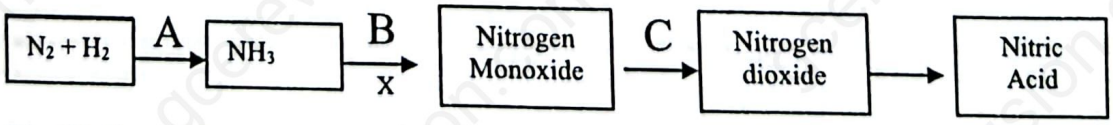
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(3 marks)

TOTAL 10 MARKS

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2 The following flow diagram shows the various stages in the manufacture of Nitric acid.



a) Identify the catalyst used in stage A

.....**(1mark)**

b) At stage B another substance X, is added

(i) Identify the substance X .....

(ii) Write a balanced equation for the reaction at B stage

.....

(3 marks)

c). How is the Nitrogen monoxide converted to nitrogen dioxide?

.....

.....

(1 mark)

d). How is the Nitrogen droxide finally converted to Nitric acid?

_____

_____ **(2 marks)**

e). The Nitric acid prepared in the laboratory, is coloured. State the colour.

.....

(1 mark)

f). Give two large scale uses of Nitric acid.

.....

..... **(2 marks)**

(TOTAL 10 marks)

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3. a) Define the following

(i). Exothermic reaction

.....

(ii). Heat of combustion.

.....

.....**(2 marks)**

b) Methanol burns as shown by the following equation

$2CH_3OH_{(aq)} + 3O_{2(g)} \longrightarrow 2CO_{2(g)} + 4H_2O_{(2)}$ $\Delta H = -726KJ/mol$

(i) Draw a labelled energy level diagram for this reaction

.....

.....

.....

.....

.....

..... **(3 marks)**

(ii). When methanol is burnt, the heat evolved is used to raise the temperature of $200cm^3$ of water by 25.4°C. Calculate the heat evolved.

.....

.....

.....

(2 marks)

c) Consider the reaction $Ag^+_{(aq)} + Cl^-_{(aq)} \longrightarrow AgCl_{(s)} \Delta H = -657$ KJ/mol

(i)Based on reactants and products only identify the reaction type. .....

(ii) What mass of silver chloride, AgCl, would be formed if 22.1KJ of energy is evolved.

.....

.....

.....

(3marks)

TOTAL = 10 marks

4. The element Rubidium, Rb, is below potassium in Group 1of the Periodic Table.

a) (i) Identify the liquid in which Rubidium is likely to be stored in the laboratory?

.....

(ii) Why is Rubidium stored in this liquid? .....

(iii) Which of the two elements is more reactive?.....

(iv) Identify one other element belonging to the same group as Rb and K,.....

(v) What is the family name of this group of elements?.....

(5 marks)

b) Write the formula of the oxide of Rubidium and state its acid base property

Formula of oxide.....

Acid base property . .....

(2 marks)

c) Write a chemical equation for the ethermal decomposition of Rubidium nitrate ($RbNO_3$).

.....(2 marks)

d) State a method by which Rubidium can be extracted from its ore.

.....(1 mark)

(Total 10 marks)

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5. a) The following formulae represent some members of two homologous series.

C_2H_4, C_2H_6,

(i) What is a homologous series?.....

.....

.....

(ii) Identify the two series

C_2H_4.....C_2H_6.....(3 marks)

b) Select from the list a compound that undergoes addition reaction with chlorine

Compound .....

Equation .....

(2 marks)

c). Briefly describe how C_2H_4 can be distinguished from C_2H_6.

.....

.....

.....

(2 marks)

d). Compound C_2H_4 can be converted to C_2H_6.

i) Identify the reagent used .....

ii) State the reaction condition .....(2 marks)

e) Give one large scale use of C_2H_4

.....

.....(1 mark)

Total 10 marks

SECTION B

ANSWER ALL 2 questions in this Section.

6. You are provided with 0.02M HNO_3 solution and a solution of Na_2CO_3. You are required to determine the concentration of the Na_2CO_3 solution by titration. The volume of HNO_3 used were tabulated as shown below.

Titration Burette reading	Approximate	1st Accurate	2nd Accurate
Initial	0.0	22.7	0.0
Final	22.7	45.3	22.6
Titre			

a) Identify the equipment that is used to

(i) Transfer $25cm^3$ of the Na_2CO_3 into a conical flask. .....

(ii) Name the equipment in which the acid is filled ..... **(2 marks)**

(iii) How is the conical flask usually cleaned before use in titration?

..... **(3 marks)**

b) Complete the table and calculate the mean

(i) Complete the table **(2 marks)**

(ii) Calculate the average titre used.

.....

.....**(2 marks)**

c) Give a suitable indicator used in the titration and state the colour change observed

Indicator .....

Colour change: Initial colour.. ..... Final colour ..... **(3 marks)**

d) Determine the concentration of the Na_2CO_3 solution in mol/dm^3

$Na_2CO_3 + 2HNO_3 \longrightarrow 2NaNO_3 + CO_2 + H_2O$

.....

.....

.....

.....

.....

..... **(3 marks)**

e) In order to prepare a dry sample of carbon dioxide, you were provided with the following. Dilute nitric acid, solid sodium carbonate, concentrated sulphuric acid, thistle funnel, 2 flat bottom flasks, delivery tube, rubber bungs, gas collecting jar.

EITHER

Draw the experimental set up that was used to prepare a pure dry sample of carbon dioxide gas

OR

Describe how the equipment and chemicals are arranged and used to prepare a pure dry sample of carbon dioxide gas.

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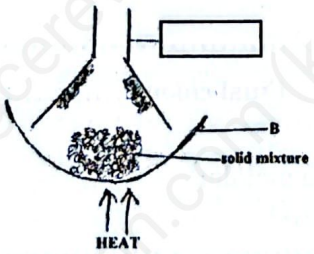
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.....**(3 marks)**

f) The following set up is used by a student to separate a solid mixtureof NH_4Cl and NaCl.



(i) Name the apparatus B.

B.....

(ii) Why was the solid mixture heated?

.....

(2 marks)

g) Why are the following usually used in the laboratory?

(i) handgloves .....

(ii) Safety googles .....

(2 marks)

(TOTAL 20 marks)

7. A series of tests were carried on 3 salts, **A B and C**. The following shows the test procedures, observation and conclusions made.

Test procedure	Observation	\Inference or conclusion
a(i)	Golden colour flame observed with A	..... 3mks
(ii) A few drops of dilute HCl are added to solid **A** in a test tube	A gas is evolved with a smell like that of burning match	.... 2mks
b(i) To $2cm^3$ of solution **B** is added a few drops of NaOH(aq)	A blue precipitate is formed	..... 1mk
(ii)To $2cm^3$ of solution B is added $BaCl_2$ followed by dilute HCl(aq)	....	SO_4^{2-} ions present 2mks
c) (i) Add NaOH(aq) to solution **C** and test with a litmus paper	....	NH_4^+ ions present 2mks
(ii)To $2cm^3$ of solution C is added acidified $AgNO_3$(aq)	White precipitate is formed	..... 1mk

d) Complete the table
Identify A, B and C

A.....

B.....

C..... **(3 marks)**

e) Complete the following table by identifying the method of collection

Gas	Draw or state the method of collection
Ammonia (NH_3)	
Sulphur dioxide (SO_2)	
Hydrogen (H_2)	3mks

c) The following data was recorded during the preparation of a standard solution of sodium carbonate
Mass of empty bottle = 5.56g
Mass of bottle plus salt = 16.10g
Calculate of the mass of the salt weighed and the amount of substance in mol
Mass of the salt weighed,

.....

.....

.....

.....

Amount of substance in mol.

.....

.....

.....

.....**(3 marks)**

(TOTAL 20 marks)

SECTION C

ANSWER 2 QUESTIONS OUT OF 3. If you attempt more than 2 questions, ONLY THE FIRST 2 will be considered.

8. Chemical substances are held by ionic, covalent or metallic bonds. Using a suitable example in each case, describe how the bonding occurs. Use appropriate diagrams to illustrate your answer.

(7,7,6 marks)

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9. You are given the following conversions. Describe giving reagents,reaction type reaction conditions and the necessary equations how each conversion is done in the laboratory
   a) Ethanol to Ethanoic acid (C_2H_5OH to CH_3COOH)
   b) Methane to Choromethane (CH_4 to CH_3Cl)
   c) Ethene to Ethane (C_2H_4 to C_2H_6)
   d) Ethanoic acid to Ethylethanoate (CH_3COOH to $CH_3COOC_2H_5$)

(5,5,5,5 marks)

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10. Sodium hydroxide is an important industrial base.
    a) Describe how sodium hydroxide is manufactured in the industry.
    b) Give two large scale uses of the chemical.

(18, 2 marks)
